

System Instructions — Supermarket Shift Management System

Running the Program

The system is provided in two formats: a source code version for development and a compiled JAR file for immediate execution.

Option 1: Running the Executable JAR

The compiled JAR is located in the `release` folder. Ensure you have **Java 21** or higher installed. Open a terminal in the project root and run:

- **With pre-loaded test data:** `java -jar release/adss2025_v01.jar 1`
- **With an empty system:** `java -jar release/adss2025_v01.jar`

Option 2: Running through Maven

You can compile and run the source code from the `dev` folder, use the following commands:

- **With pre-loaded test data:**
`mvn compile exec:java -Dexec.mainClass="dev.presentation.Main" -Dexec.args="1"`
- **Empty system:**
`mvn compile exec:java -Dexec.mainClass="dev.presentation.Main"`

Pre-loaded Test Data

When running with argument `1`, the system initializes with the following data:

Users

ID	Password	Role	Name
100000001	sarah123	HR Manager + Shift Manager	Sarah Cohen
100000002	yossi123	Cashier	Yossi Levi

ID	Password	Role	Name
100000003	dana1234	Warehouse + Cashier	Dana Mizrahi
100000004	ron12345	Shift Manager + Cashier	Ron Shapiro

Pre-loaded Shifts

Shift ID	Date	Type	Assigned Employees
SHIFT_001	next Tuesday	Morning	Ron (Manager), Yossi (Cashier), Ron (Cashier), Dana (Warehouse)
SHIFT_002	next Wednesday	Morning	Sarah (Manager), Dana (Cashier)
SHIFT_003	next Thursday	Morning	Ron (Manager), Yossi (Cashier), Dana (Warehouse)
SHIFT_004	15/01/2026	Morning	Ron (Manager), Yossi (Cashier), Dana (Warehouse)

Pre-loaded Constraints

- Yossi Levi has submitted availability for Tuesday and Thursday morning shifts.

Pre-loaded Override Request

- Sarah Cohen sent an override request to assign Yossi Levi as Cashier on Wednesday's morning shift.

Pre-loaded Archive

- A past morning shift from January 15, 2026 (SHIFT_004) has been archived and is available for viewing via the **View History** option.
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Login

Upon launching, the system prompts for:

1. **Employee ID** — a 9-digit number
2. **Password** — at least 6 characters

The system automatically detects whether the user is an HR Manager or a regular Employee and routes them to the appropriate menu.

HR Manager Menu

After logging in as an HR Manager, the following options are available:

1. Shift Definition

- **Create Shift** — Select a day of the week and shift type (Morning/Evening) to create a new shift.
- **Set Requirements** — Select a shift, then specify how many employees of a given role (Cashier, Warehouse, Shift Manager) are required.

2. Shift Filling

- **View next week's schedule** — Displays all active shifts with their current assignments.
- **Assign employee** — Select a shift, role, and employee. If no available employees exist, the system offers to send an override request.
- **Unassign employee** — Remove an employee from a specific role in a shift.
- **View employee constraints** — Enter an employee ID to view their weekly availability submission.
- **View employee preferences** — Enter an employee ID to view their weekly preferences submission.

3. Employee Management

- **Hire employee** — Enter all employee details (ID, password, name, bank account, start date, employment type, salary type, salary, vacation days, overtime preference, day off, double shift preference, certifications).
- **Dismiss employee** — Enter employee ID to deactivate them.
- **Rehire employee** — Enter employee ID and new password to reactivate a dismissed employee.
- **View employee data** — Enter employee ID to view full profile.
- **Edit employee data** — Sub-menu with options:
 - Financial Details (bank account, salary type, salary)
 - Employment Details (employment type, vacation days, start date)
 - Name
 - Certifications (add or remove)

- Settings (day off, double shift preference, overtime preference)

4. Request Handling

- **View requests** — Displays all override requests sent by this HR manager. Select a request to view details. If the request has been approved by the employee, the HR manager can proceed to assign with override.

5. View History

- Enter a date and shift type to view details of a past (archived) shift.

6. Logout

- Logs out the current user and returns to the login screen.
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Employee Menu

After logging in as a regular employee, the following options are available:

1. Availability

- **View constraints** — Displays the employee's current weekly availability submission.
- **Set constraints** — For each day and shift of the upcoming week, the employee indicates whether they are available. Submitting an empty response clears all previous submissions.
- **View preferences** — Displays the employee's current weekly preferences.
- **Set preferences** — For each day and shift, the employee indicates preferred shifts. Note: preferences can only be set for shifts where a constraint (availability) was already submitted.

2. Requests

- **View requests** — Displays all override requests received by this employee. The employee can select a request and choose to approve or reject it.

3. My Profile

- **View my details** — Displays the employee's full profile (ID, name, status, bank account, start date, employment type, salary, vacation days, day off, certifications).
- **Edit settings** — Allows the employee to update their day off, double shift preference, and overtime preference.

4. Logout

- Logs out the current user and returns to the login screen.
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Business Rules

- There can only be **one HR Manager** in the system.
- Every shift must have at least **one Shift Manager**.
- An employee **cannot be assigned** to a shift on their day off.
- An employee can only be assigned to a shift if they submitted availability (constraint) for that slot — unless an override request is approved.
- A Shift Manager can hold **up to 2 roles** in the same shift.
- Preferences can only be submitted for slots where availability was already submitted.
- Override requests must be **approved by the employee** before the HR manager can assign with override.